To achieve command-line handling in modules, we first need to declare global variables and use the *module_param()* macro, which is defined in the *<linux/moduleparam.h>* header file. There is also the *MODULE_PARM_DESC()* macro which provides descriptions about arguments. Without going into lengthy theoretical discussions, let us write the code:

```
#include <linux/kernel.h>
#include <linux/module.h>
#include <linux/moduleparam.h>

static char *name = "Narendra Kangralkar";
static long roll_no = 1234;
static int total_subjects = 5;
static int marks[5] = {80, 75, 83, 95, 87};

module_param(name, charp, 0);
MODULE_PARM_DESC(name, "Name of the a student");

module_param(roll_no, long, 0);
MODULE_PARM_DESC(rool_no, "Roll number of a student");

module_param(total_subjects, int, 0);
MODULE_PARM_DESC(total_subjects, "Total number of subjects");

module_param_array(marks, int, &total_subjects, 0);
MODULE_PARM_DESC(marks, "Subjectwise marks of a student");

static int __init param_init(void)
{
    static int i;

    printk(KERN_INFO "Name             : %s\n", name);
    printk(KERN_INFO "Roll no          : %ld\n", roll_no);
    printk(KERN_INFO "Subjectwise marks ");

    for (i = 0; i < total_subjects; ++i) {
      printk(KERN_INFO "Subject-%d = %d\n", i + 1,
marks[i]);
    }

    return 0;
}

static void __exit param_exit(void)
{
    /* Don't do anything */
}

module_init(param_init);
module_exit(param_exit);

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Narendra Kangralkar.");
MODULE_DESCRIPTION("Module with command line arguments.");
MODULE_VERSION("1.0");
```

After compilation, first insert the module without any arguments, which display the default values of the variable. But after providing command-line arguments, default values will be overridden. The output below illustrates this:

```
[root]# insmod parameters.ko

[root]# dmesg
Name             : Narendra Kangralkar
Roll no          : 1234
Subjectwise marks
Subject-1 = 80
Subject-2 = 75
Subject-3 = 83
Subject-4 = 95
Subject-5 = 87

[root]# rmmod parameters
```

Now, let us reload module with command-line arguments and verify the output.

```
[root]# insmod ./parameters.ko name="Mickey" roll_no=1001
marks=10,20,30,40,50

[root]# dmesg
Name             : Mickey
Roll no          : 1001
Subjectwise marks
Subject-1 = 10
Subject-2 = 20
Subject-3 = 30
Subject-4 = 40
Subject-5 = 50
```

If you want to learn more about modules, the Linux kernel's source code is the best place to do so. You can download the latest source code from *https://www.kernel.org/*. Additionally, there are a few good books available in the market like 'Linux Kernel Development' (3rd Edition) by Robert Love and 'Linux Device Drivers' (3rd Edition). You can also download the free book from http://lwn.net/Kernel/LDD3/. END

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